



University
of Glasgow

W2-02: MapReduce

COMPSCI4064 (H) and COMPSCI5088 (M)

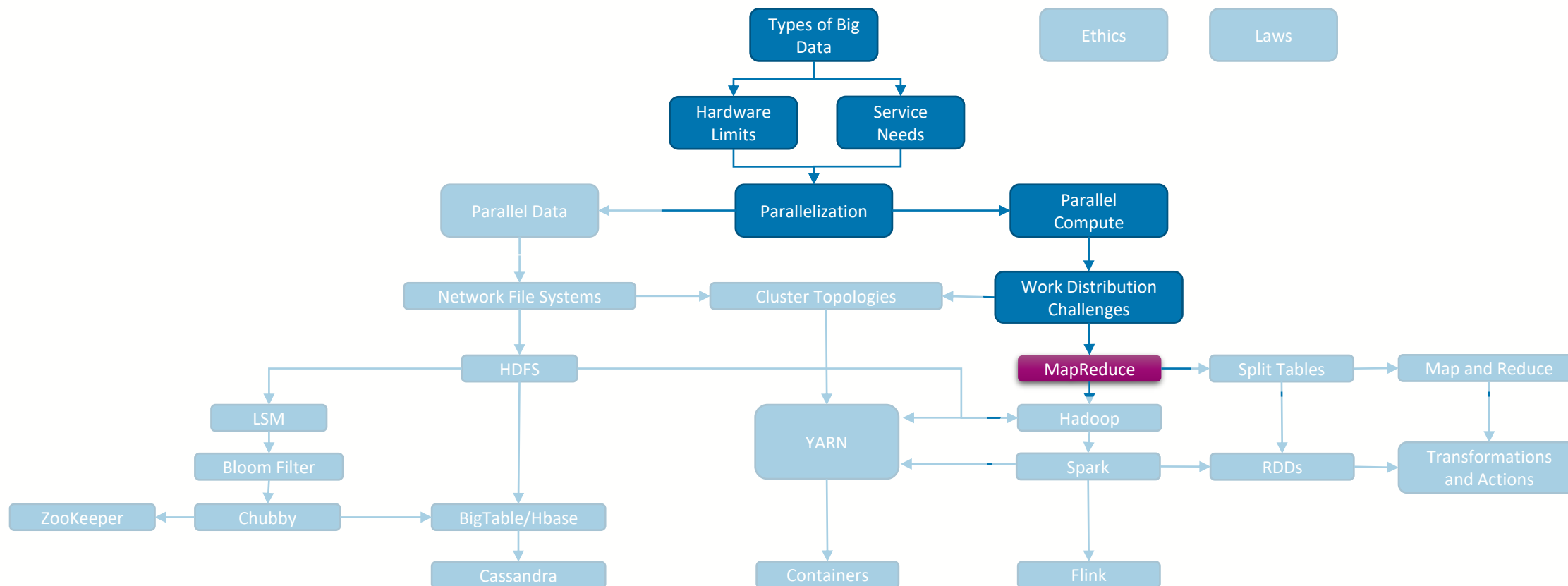
Dr. Richard McCreddie

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Where are we?





Work Distribution Challenges

- In the previous lecture we discussed why work distribution is challenging:
 - **Overheads** – additional costs due to moving and aggregating data
 - **Load Balancing** – making sure that all of our machines have work to do
 - **Orchestration** – physically distributing work and handling failures
- The first major solution to *all* of these problems is known as **MapReduce**
 - (there were up to that point many solutions that solved a subset of the above)

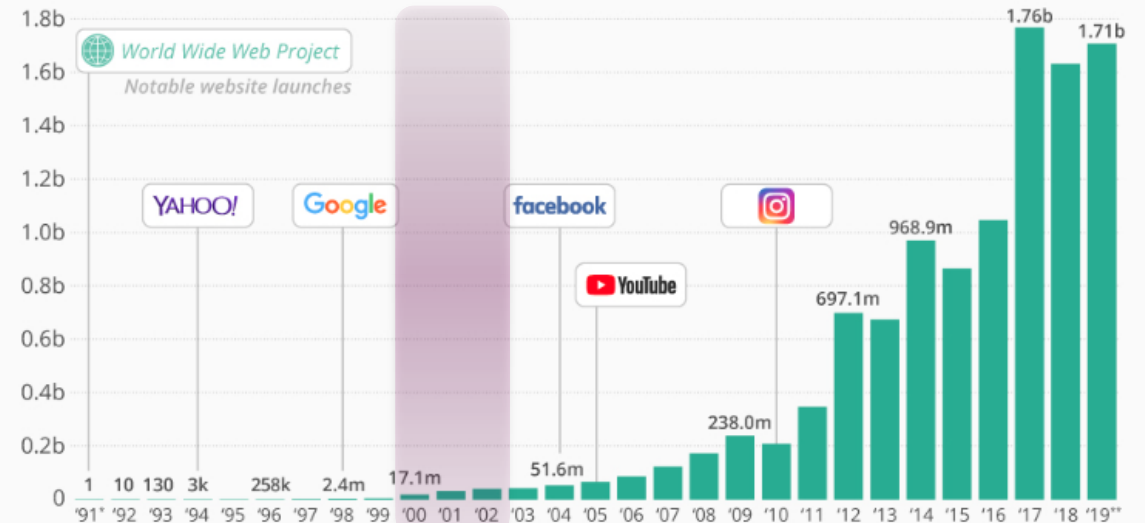


Setting the Scene...

- It's the early 2000's, and a relatively small search company called **Google** is trying to scale up the amount of the Internet their search engine covers
- They have crawler programs that are following links on the Web to find new web pages, but the number of new websites is growing rapidly...

How Many Websites Are There?

Number of websites online from 1991 to 2019



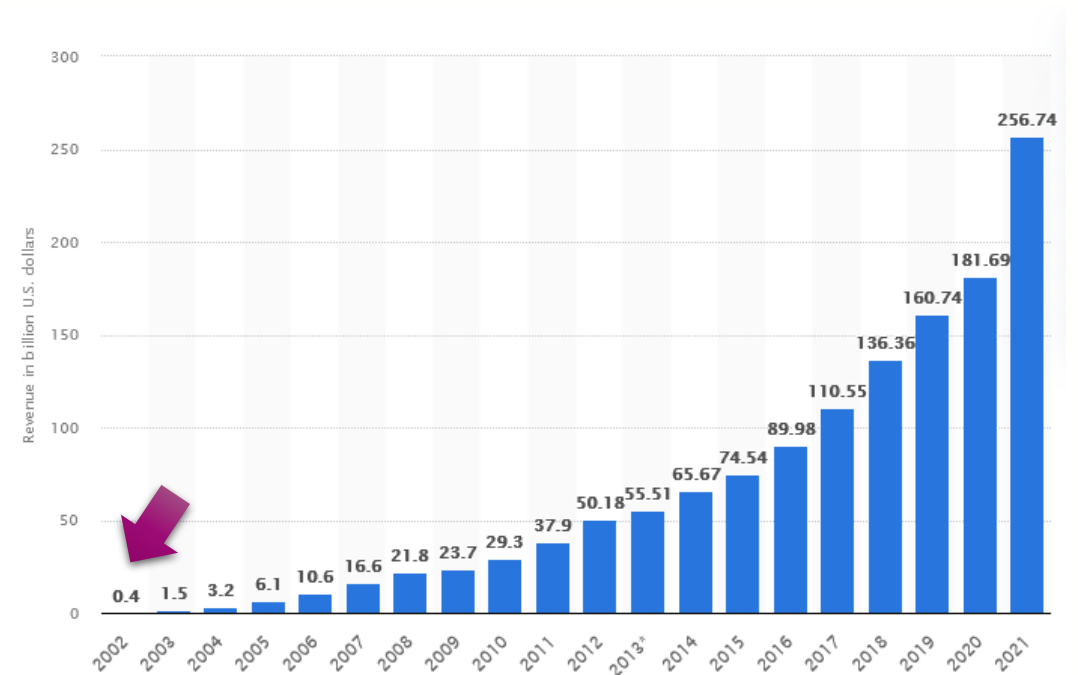
"Website" is defined as a unique hostname, i.e. a name which can be resolved, using a name server, into an IP Address.

* As of August 1, 1991
** As of August 19, 2019 at 10:22 CET
@StatistaCharts Source: Internet Live Stats



Setting the Scene...

- At this time Google is growing, but is still quite small
- Buying ever bigger mainframes is not practical given the size of the Web, so they start working on a **horizontal scaling** solution...



Details: Worldwide; Google; 2002 to 2021

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Google Revenue over time (Billions USD)



Continuous Index Growth

- Between 2000 and 2004, the Google Index **grew over 20x**
- In 2004 Google submitted a patent for a new invention and also published an associated research paper:

MapReduce: simplified data processing on large clusters

[J Dean](#), [S Ghemawat](#) - Communications of the ACM, 2008 - dl.acm.org

... the performance of **MapReduce** on two computations running on a **large cluster** of machines. One computation searches through approximately one terabyte of **data** looking for a particu...

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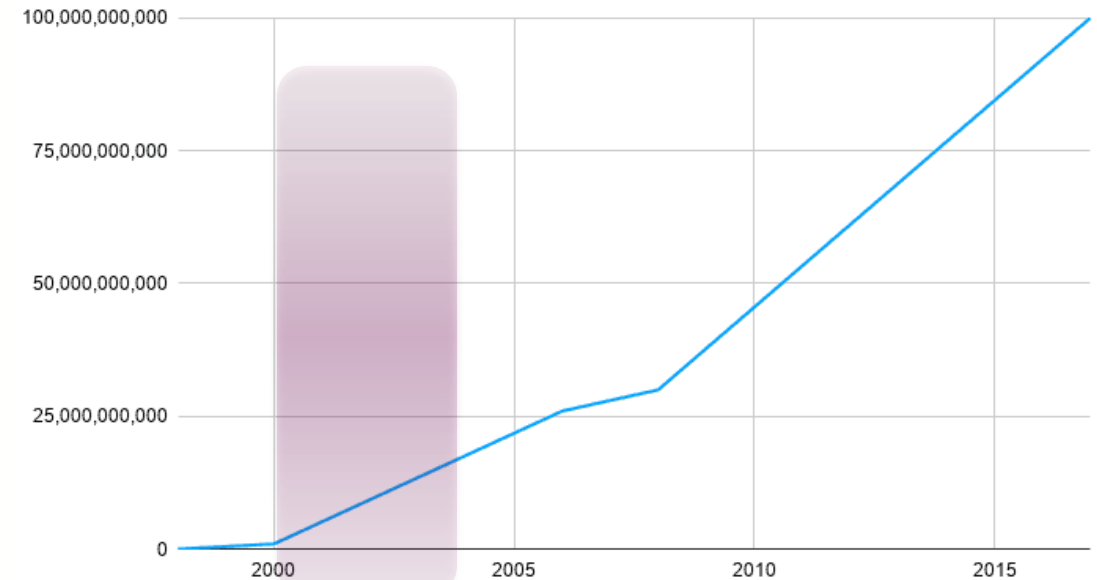
[HTML] MapReduce: Simplified data processing on large clusters

[J Dean](#), [S Ghemawat](#) - 2004 - usenix.org

... Our implementation of **MapReduce** runs on a **large cluster** of ... **MapReduce** computation **processes** many terabytes of **data** on ... the system **easy** to use: hundreds of **MapReduce** programs ...

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Indexed pages reported by Google



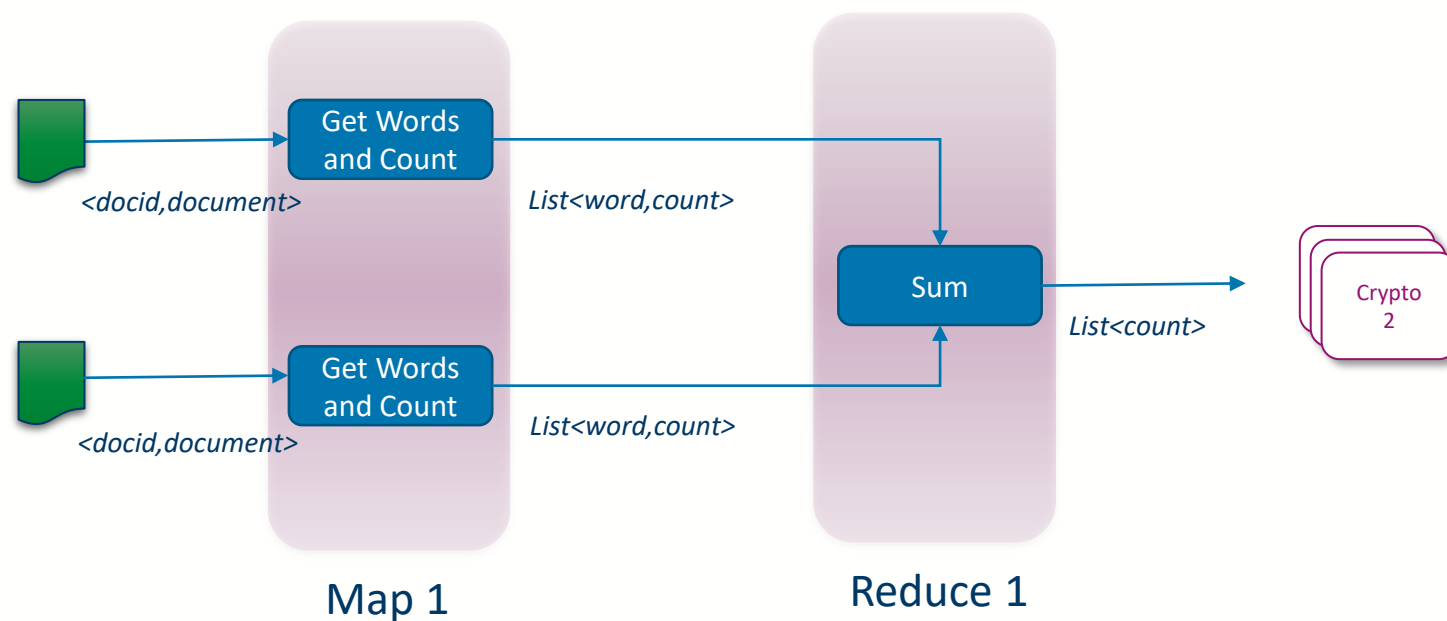


MapReduce

- This new ‘invention’ was called **MapReduce**
- The core of MapReduce is very simple:
 - The majority of distributed compute workloads can be defined as the combination of two operations
 - **Map**(Key1, Value1) $\longrightarrow$ List[<Key2, Value2>]
 - **Reduce**(Key2, List[Value2]) $\longrightarrow$ List<Value2>
- Google’s answer to Big Data circa 2004: if we build a framework that only supports these two operations, it makes solving all of those workload distribution challenges much easier while still supporting our company’s needs

Word Counting Example Again

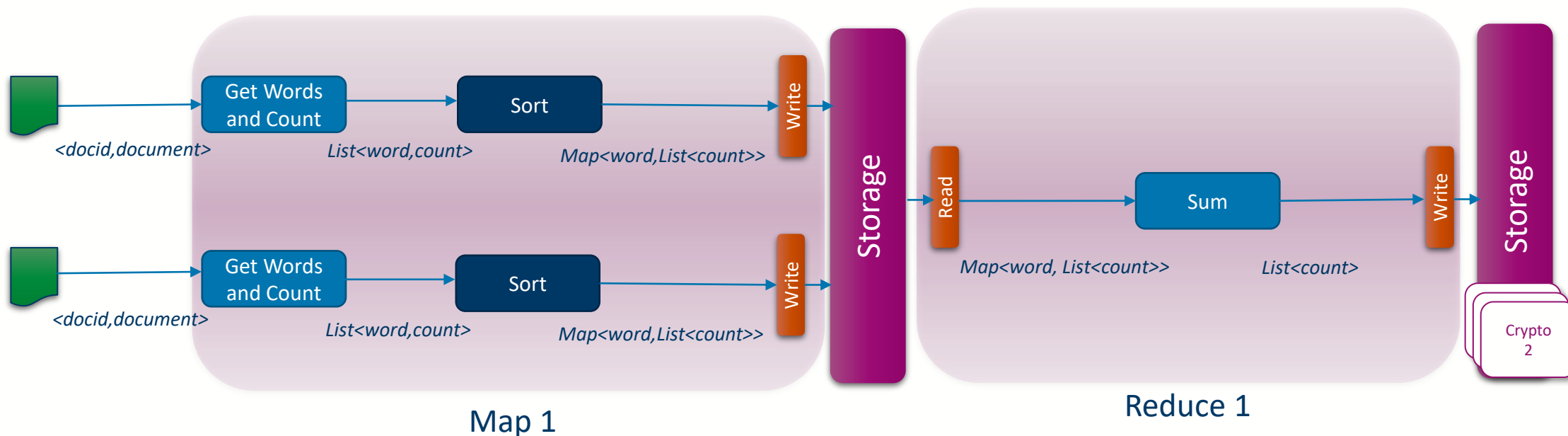
- How would the parallel word counting example we looked at previously work under MapReduce?



This is the basic conceptual view of the word count application
... but wait, are not the inputs to the reducer wrong?

MapReduce Sort and Read/Write

- MapReduce is designed for parallel fault-tolerant computation, what this means is that there are intermediate stages between maps and reducers, where data gets re-organized and then persistently stored





MapReduce Popularity

- The power of MapReduce was its **simplicity**
 - Has only two types of operations
 - The data transfer flow is clear and easy to understand
- Most Big Data Applications can be written as MapReduce Jobs
 - **Extraction/Filtering**: I have a large silo of data and I need to extract a subset of it matching some criteria - **Map**
 - **Transformation**: I have a function that I want to apply to each item in my data and save the outcome - **Map**
 - **Aggregation**: I want to merge (and typically concurrently transform) a set of data items - **Reduce**



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